

PROBLEM 1

Suppose M_1 is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

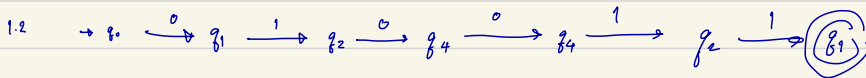
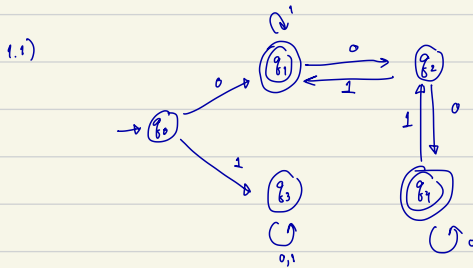
	0	1
q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_3
q_4	q_4	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

1.1) Draw a state diagram of machine M_1 .

1.2) Write the computation path of M_1 on the string 010011. Is this string accepted by M_1 ?

1.3) Find all the strings of length not greater than 4 accepted by M_1 .



Accepted

- 1.3
- length 1 : 0
- length 2 : 01
- length 3 : 011, 001
- length 4 : 0111, 0101, 0011, 0000

PROBLEM 2

Let L_1 and L_2 be the following languages on the alphabet $\Sigma = \{0,1\}$:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\}$$

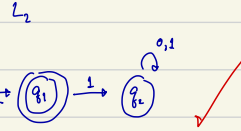
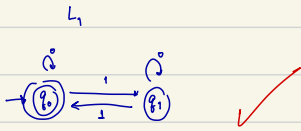
$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}$$

2.1 Design (deterministic) finite automata M_1 and M_2 that recognize L_1 and L_2 , respectively.

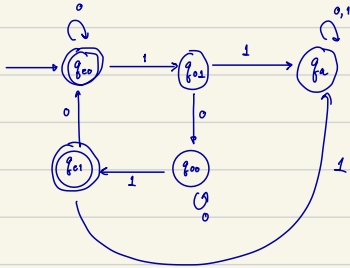
2.2 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cap L_2$.

2.3 Construct a product automaton of M_1 and M_2 that recognizes the language $L_1 \cup L_2$.

2.1

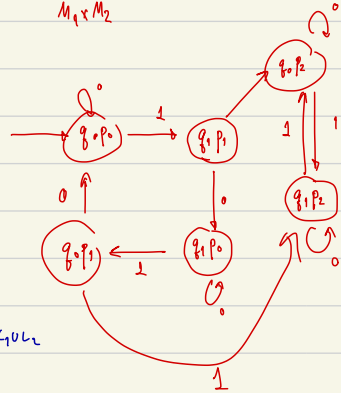


2.2

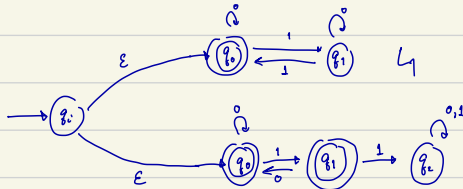


$L_1 \cap L_2$

$M_1 \times M_2$

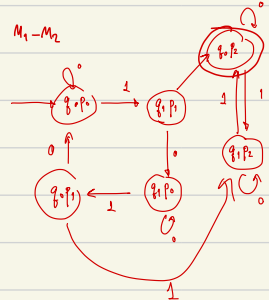


2.3

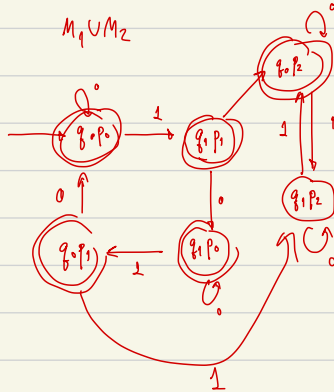
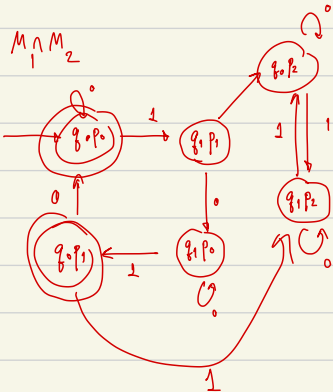
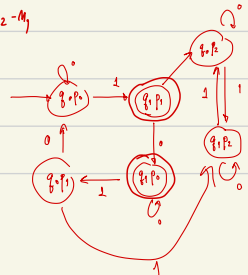


$L_1 \cup L_2$

$M_1 - M_2$



$M_2 - M_1$



PROBLEM 3

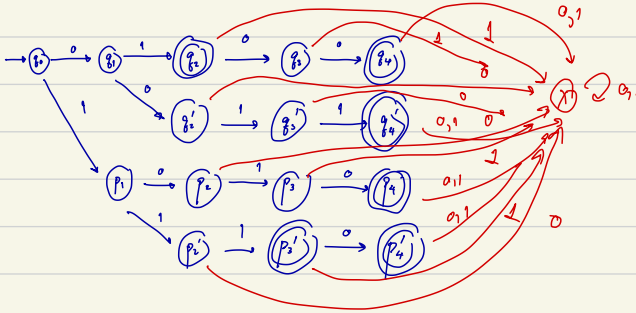
3.1 Design a (deterministic) finite automaton that recognizes the following language L on the alphabet $\Sigma = \{0,1\}$:

$$L = \{0011, 1010, 1110, 0100, 11101\}$$

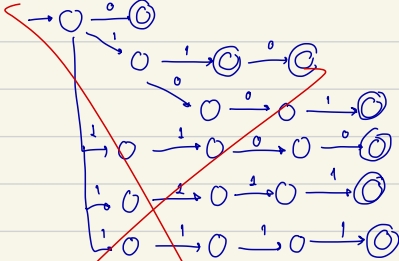
3.2 Design a (deterministic) finite automaton that recognizes the following language L' on the alphabet $\Sigma = \{0,1\}$:

$$L' = \{w \in \Sigma^* \mid w \text{ is a binary representation of a non-negative integer divisible by } 3\}$$

3.1



3.2



3.2

$0 \rightarrow 0 \rightarrow 0, 00, 000, \dots$
 $3 \rightarrow 11 \rightarrow 1, 011, 0011, \dots$
 $6 \rightarrow 110$
 $4 \rightarrow 1001$
 $12 \rightarrow 1100$
 $15 \rightarrow 1111$
 $18 \rightarrow 10010$

ε

$$L_0 ; x \bmod 3 = 0$$

$$L_1 ; x \bmod 3 = 1$$

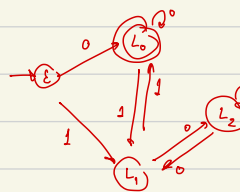
$$L_2 ; x \bmod 3 = 2$$

$$\text{If } w \in L_0 \rightarrow V(w) = 3x \rightarrow V(w_0) = 2 \cdot V(w) = 2 \cdot 3x \rightarrow w_0 \in L_0$$

$$\text{If } w \in L_1 \rightarrow V(w) = 3x+1 \rightarrow V(w_0) = 2 \cdot V(w) = 2 \cdot (3x+1) = 2 \cdot 3x+2 \rightarrow w_0 \in L_2$$

$$\text{If } w \in L_2 \rightarrow V(w) = 3x+2 \rightarrow V(w_0) = 2 \cdot V(w) = 2 \cdot (3x+2) = 2 \cdot 3x+4 \rightarrow w_0 \in L_1$$

$$\text{If } \varepsilon \rightarrow L_0$$



PROBLEM 4

Suppose M_2 is a nondeterministic finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{p_0, p_1, p_2, p_3, p_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

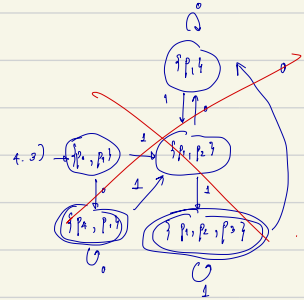
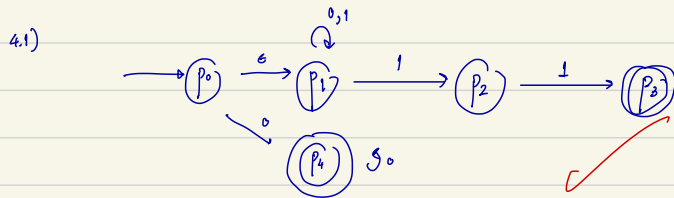
	ϵ	0	1
p_0	$\{p_1\}$	$\{p_4\}$	$\emptyset$
p_1	$\emptyset$	$\{p_1\}$	$\{p_1, p_2\}$
p_2	$\emptyset$	$\emptyset$	$\{p_3\}$
p_3	$\emptyset$	$\emptyset$	$\emptyset$
p_4	$\emptyset$	$\{p_4\}$	$\emptyset$

- p_0 is the initial state,
- $F = \{p_3, p_4\}$

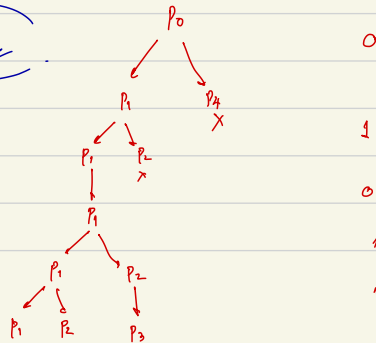
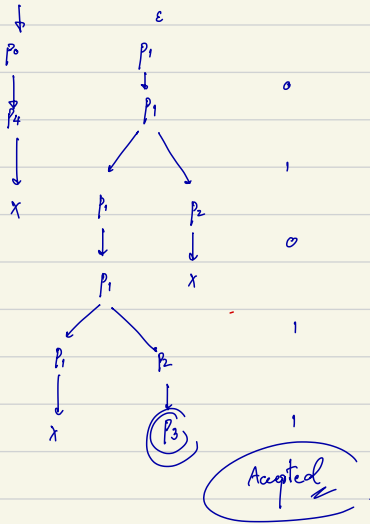
4.1) Draw a state diagram of machine M_2 .

4.2) Write a computation tree of M_2 on the string 01011. Is this string accepted by M_2 ?

4.3) Apply subset construction to construct a (deterministic) finite automaton that is equivalent to M_2 .



4.2)

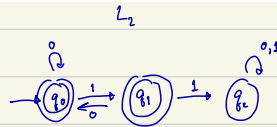
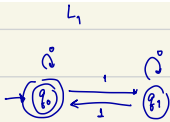


PROBLEM 5

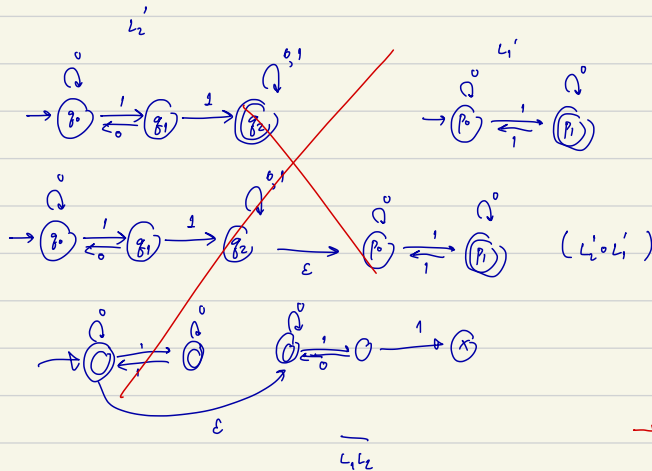
Construct a nondeterministic finite automaton recognizing the following language $\Sigma^* - (L_1 L_2)$ where L_1 and L_2 are languages on the alphabet $\Sigma = \{0,1\}$ given as follows:

$$L_1 = \{w \in \Sigma^* \mid w \text{ contains an even number of occurrences of } 1\}$$

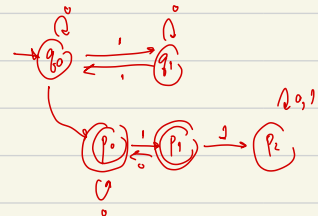
$$L_2 = \{w \in \Sigma^* \mid w \text{ contains no adjacent occurrences of } 1\}$$



Find $\Sigma^* - (L_1 L_2) \rightarrow (L_1 L_2)' \rightarrow L_2' \circ L_1'$



$M_1 M_2$



It's not in DFA.

